

PAPER AI and Quantum Computing for Fusion: ML Surrogates, Real-Time Control, and a Resource-Honest Quantum Frontier

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Abstract

A nuclear-regulated fusion generator must be controlled across ten orders of magnitude in time — microsecond magnet protection through month-scale fuel logistics — and every automated actuation must be bounded, auditable, and reproducible. We present the umbrella account of the Kronos computational and control programme and hold its three technology-readiness tiers strictly apart. *Physics-informed machine learning is deployable now, in the control loop, when its output is deterministically clamped*: a model-predictive controller acts inside a certified control-barrier-function (CBF) safety projection whose forward-invariance we prove analytically and realise as a small quadratic program with a solver-free closed form; over 100 Monte-Carlo trajectories the clamp admits 0 safe-set violations against 50 unclamped, holding the peak normalized pressure on the limit ($\beta_N \leq 3.5$, $h_{\min} = +0.000$). A physics-informed neural network reconstructs the negative-triangularity Grad-Shafranov equilibrium to 0.139% RMS ψ against a free-boundary solver (2877 $\times$ faster), a graph network fuses degraded diagnostics, and a fused strain-rate+magnetization quench-precursor detector reaches an area under the receiver-operating characteristic of 0.992. The models are made predictive by a four-twin, 50–100 ms shadow, carry calibrated uncertainty, and are admitted to control authority only through pre-registered acceptance inequalities. *Quantum sensing is a near-term diagnostics enhancement; quantum computation is, for fusion, an offline calibration tier with no resource crossover this decade*: the linear kinetic operator needs 11–19 logical qubits (early fault-tolerant), the 6-D kinetic and exact-alloy problems need ~ 44 and ~ 238 logical qubits, and we report a measured quantum-machine-learning *negative* (classifier AUC falls $0.817 \rightarrow 0.691$ as the feature map widens $4 \rightarrow 8$ qubits). The entire campaign runs on the NVIDIA GPU HPC campaign against frozen reference codes. The contribution is a certifiable, resource-honest *system and method*, not new plasma physics; every headline quantity is a frozen anchor in the de-risking register and crosses the Kronos 2026 series. Canonical anchors are the Hyperion breeder ($Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $\delta = -0.30$, centrepost 16.84 T) and the D-³He tandem-mirror burner ($Q_E = 1.318$, plug 26.49 T) — two distinct magnets.

Keywords: digital twin; physics-informed machine learning; neural operators; control barrier functions; model-predictive control; safe reinforcement learning; quantum computing; quantum sensing; disruption prediction

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1 Introduction

1.1 *Fusion is a control problem*

A compact, high-power-density fusion generator is as much a control problem as a physics problem. The Hyperion breeder holds a plasma current of $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak on-conductor field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime chosen for good confinement without the edge-localised-mode cycle of conventional H-mode [5, 6]. In a device this compact the characteristic times of vertical instability, thermal transients and magnet events are short; in a nuclear system every actuation carries a safety consequence [2, 3, 4]. Autonomy is therefore not a convenience but a design requirement, and an autonomous controller is only as trustworthy as the model beneath it and the floor beneath that model.

Feedback control of tokamak equilibria and profiles is a mature discipline — magnetic reconstruction feeds real-time equilibrium [7] and shape controllers [8], and physics-based profile simulators run inside the control cycle [9]. More recently, learned controllers have entered the loop: Degraeve *et al.* trained a deep reinforcement-learning (RL) agent to command TCV coil voltages directly from magnetics, including negative-triangularity shapes [10]; Seo *et al.* used RL to steer DIII-D clear of a tearing instability [11]; and deep networks forecast disruptions across machines [12]. These advances share a thread: control quality is bounded by the speed and fidelity of the underlying model, and none of them, by itself, supplies the certified last line of defence a regulated plant requires.

1.2 *Three readiness tiers, held apart*

Machine learning and quantum technologies are routinely invoked for fusion without a clear statement of what is deployable now versus what is a decade away. This paper is the umbrella of the Kronos computational programme and holds three tiers strictly apart, reporting each with its evidence and its misses:

- **Tier 1 — physics-informed AI, deployable now if clamped.** Learned surrogates and optimisation-based controllers run in the loop *because* their error is bounded and deterministically clamped by a certified safety projection. This is the load-bearing contribution.
- **Tier 2 — quantum sensing, near-term diagnostics.** Magnetometry at or near the quantum limit sharpens the state estimate that feeds the loop; it enhances the sensing layer rather than replacing it.
- **Tier 3 — quantum computation, an offline calibration tier.** For fusion there is no resource crossover this decade. We give dated fault-tolerant resource estimates and report a measured quantum-machine-learning *negative* rather than reframing it.

1.3 *Contribution and the series it heads*

The contribution is a *system and a method*: (i) an eight-layer, four-twin autonomous control architecture made predictive by learned neural operators and carrying calibrated uncertainty end to end (§2); (ii) a governing-equation account of the safety-projected controller — the model-predictive-control (MPC) quadratic program (QP) bounded by a control-barrier-function projection with an analytic forward-invariance proof (§3); (iii) the learned-surrogate and uncertainty machinery — physics-informed and graph learning, ensemble assimilation, conformal and out-of-distribution (OOD) gating (§4); (iv) a reproducible computational methodology on the NVIDIA GPU HPC campaign (§5); and (v) a resource-honest quantum frontier with a dated roadmap (§6). Every quantitative claim is a frozen anchor in the Kronos Physics De-Risking Register [1]; serial identifiers (e.g. KX-L1-A13) are given inline so each number is auditable, and each sub-topic has a dedicated companion paper in the 2026 series, cross-cited by reserved DOI in §10. The contribution is not new plasma physics.

2 The autonomous control architecture

2.1 *Eight layers and the last line of defence*

The stack spans eight numbered layers, from bare-metal and field-programmable-gate-array (FPGA) hardware at the foundation (L0), through infrastructure and data (L1–L2), the AI capabilities and models (L3), workflows and copilots (L4–L5), to the experience and ecosystem layers (L6–L7) that face operators and markets (figure 1). The identical architecture drives both machines: only the physical plant, the actuators, and the physics models differ between the

Hyperion breeder and the D-³He tandem-mirror burner. The organising principle is that *no AI output reaches an actuator without passing a deterministic, hardware-enforced clamp*. The learned layers pursue performance; a proven boundary beneath them guarantees the machine stays in its safe envelope even when a model is wrong. That boundary — deterministic rules plus a hardware failsafe, exercised in 0.055 μ s at the magnet dump — needs neither the network nor the cloud, so loss of the AI degrades the plant to a safe state rather than an unsafe one.

(Schematic.) Eight-layer control stack and the four coupled twins

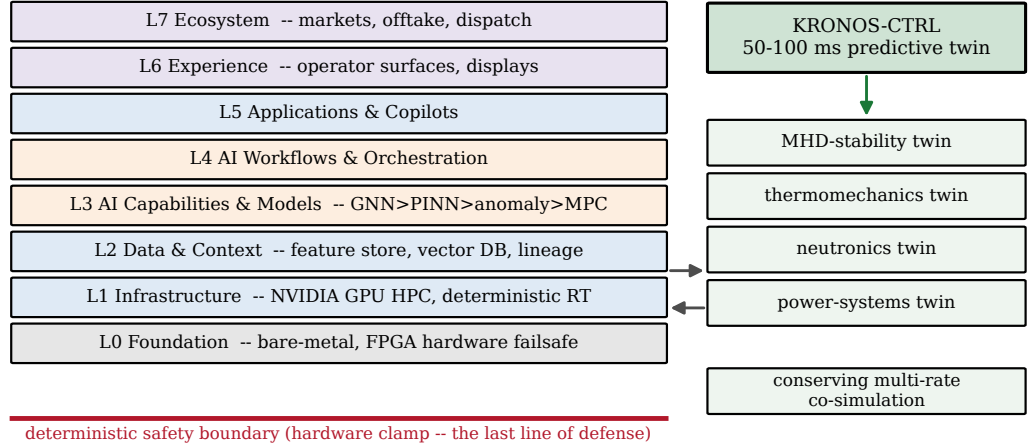


Figure 1. (Schematic.) The eight-layer control stack (L0–L7), the four coupled twins, and the KRONOS-CTRL 50–100 ms predictive twin. A deterministic safety boundary sits below the learned layers: the hardware clamp is the last line of defence. The figure encodes structure only; no data values are shown.

2.2 The four coupled twins and the predictive shadow

The plant is shadowed by four digital twins, each owning a physical domain and exposing a small set of interface quantities: an MHD-stability twin (free-boundary equilibrium and ideal-MHD margins), a thermomechanics twin (the heat path and the structure), a neutronics twin (the 14 MeV source, breeding, activation and dose [42]), and a power-systems twin (magnets, cryogenics, quench state). The twins are coupled through a conserving multi-rate co-simulation and made predictive by learned flux and equilibrium neural operators, so the controller acts on where the plasma *will* be, not where it was; the shadow lead-time is held at $\tau_{\text{lead}} \geq 50$ ms. The full formalism — coupling boundary quantities, the energy/flux conservation gates, the ensemble-Kalman assimilation and the demonstrated magnet subsystem — is the subject of the companion four-twins paper; here we state the pieces the umbrella needs and give the equations in §3–§4.

2.3 The model pipeline: $GNN \rightarrow PINN \rightarrow anomaly \rightarrow MPC$

The real-time loop (figure 2) chains four learned or optimisation-based stages inside a sub-millisecond budget. A graph neural network (GNN) reconstructs the plant state from the diagnostic set and imputes dropped channels; a physics-informed neural network (PINN) supplies a fast, differentiable equilibrium and the neural-operator transport surrogate supplies the fluxes; an anomaly-ensemble flags precursors; and an MPC reference governor computes the nominal actuation. Every candidate action then passes the CBF safety projection before it reaches an actuator. The estimate that enters the loop is calibrated: a state distribution, not a point, so the safety layer can read how far to trust the model.

2.4 Maturity-gated actuation authority

Control authority is not binary. Each learned component is admitted to a graded level of authority — shadow-only, advisory, supervised, bounded, full — and rises only as it clears successive, pre-registered acceptance inequalities (figure 3); the mapping from evidence to authority is explicit, so a model that has passed only its surrogate-error monitor may inform but not command. This

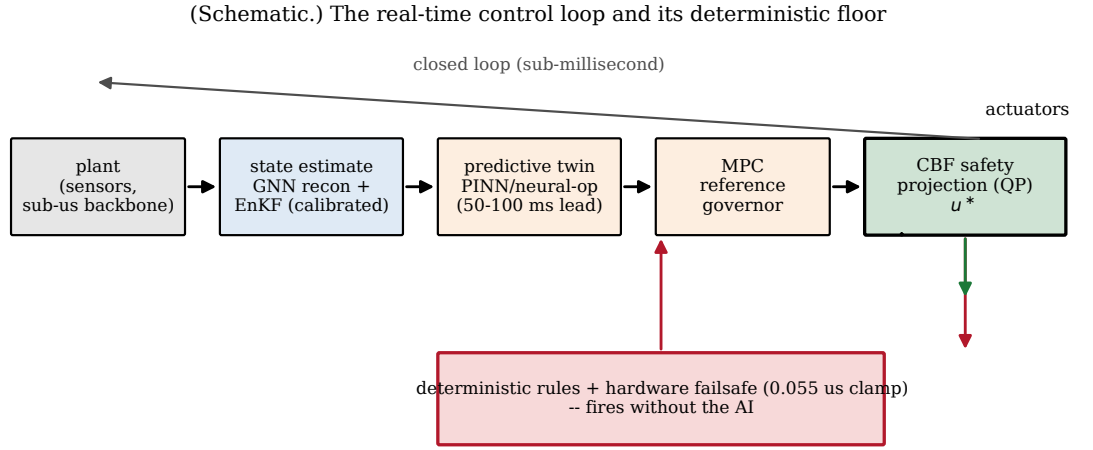


Figure 2. (Schematic.) The sub-millisecond real-time control loop: calibrated state estimate $\rightarrow$ predictive twin $\rightarrow$ MPC reference governor $\rightarrow$ CBF safety projection $\rightarrow$ actuator, with a deterministic rules-plus-hardware floor that fires without the AI. Structure only; no data plotted.

binds runtime authority to verification credibility and makes the path from a demonstrated subsystem to a fully autonomous plant a defined, testable sequence rather than an open question. The gate is the runtime expression of the assurance case (§10).

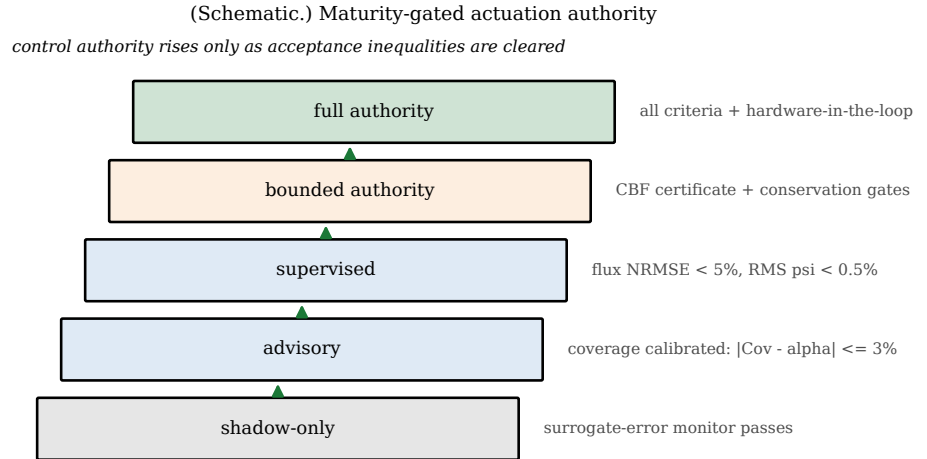


Figure 3. (Schematic.) Maturity-gated authority. Each rung admits a higher level of control authority and is unlocked by the pre-registered acceptance inequalities of §4. The criteria are the real, frozen bars; the authority ladder is a schematic of the gating policy.

3 Governing equations and the safety-projected controller

We now give the mathematics under the acting layer: the controller as an optimisation problem, and the deterministic clamp as a control-barrier projection with a proof of forward invariance.

3.1 Receding-horizon model-predictive control

Write the plant state $\mathbf{x} \in \mathbb{R}^n$, actuation $\mathbf{u} \in \mathbb{R}^m$, and the differentiable twin $\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k)$ (the learned operators of §4). The acting layer solves, at each control tick, the constrained receding-horizon problem

$$\min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} \left[\|\mathbf{x}_k - \mathbf{x}^*\|_Q^2 + \|\mathbf{u}_k\|_R^2 \right] \quad \text{s.t.} \quad \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \quad \mathbf{x}_k \in \mathcal{X}, \quad \mathbf{u}_k \in \mathcal{U}, \quad (1)$$

with tracking weight $Q \succeq 0$ and effort weight $R \succ 0$, target operating point $\mathbf{x}^*$, and admissible sets $\mathcal{X}, \mathcal{U}$. About the current shadow state the twin is linearised by reverse-mode automatic

differentiation, $\mathbf{x}_{k+1} \approx f(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{J}_k(\mathbf{x} - \mathbf{x}_k)$ with $\mathbf{J}_k = \partial f / \partial \mathbf{x}|_{\mathbf{x}_k}$, so (1) reduces to a convex QP the controller solves within the control period [29]. A reference governor shapes $\mathbf{x}^*$ to keep the horizon feasible. Equation (1) is the performance layer; on its own it offers no guarantee, because f is learned and imperfect. The guarantee is supplied by the projection below.

3.2 The control-barrier safety projection

Take the control-affine continuous-time model $\dot{\mathbf{x}} = f_c(\mathbf{x}) + g_c(\mathbf{x})\mathbf{u}$ and a safe set encoded as the zero-superlevel set of a smooth function h ,

$$\mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}, \quad \partial\mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) = 0\}. \quad (2)$$

h is a *control barrier function* [27, 28] for (2) if there is an extended class- $\mathcal{K}$ function α such that

$$\sup_{\mathbf{u} \in \mathcal{U}} \left[\underbrace{L_{f_c} h(\mathbf{x})}_{\nabla h \cdot f_c} + \underbrace{L_{g_c} h(\mathbf{x}) \mathbf{u}}_{\nabla h \cdot g_c} \right] \geq -\alpha(h(\mathbf{x})) \quad \forall \mathbf{x} \in \mathcal{C}. \quad (3)$$

The deterministic clamp takes the MPC's proposed action $\mathbf{u}_{\text{nom}}$ from (1) and returns the *nearest* admissible action that satisfies (3):

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathcal{U}} \frac{1}{2} \|\mathbf{u} - \mathbf{u}_{\text{nom}}\|^2 \quad \text{s.t.} \quad L_{f_c} h(\mathbf{x}) + L_{g_c} h(\mathbf{x}) \mathbf{u} \geq -\alpha(h(\mathbf{x})) + \kappa_d, \quad (4)$$

a strictly convex QP with a single (barrier) inequality, where $\kappa_d \geq \sup_{\|\mathbf{d}\| \leq \bar{d}} |\nabla h \cdot \mathbf{d}|$ robustly tightens the constraint against a bounded disturbance $\mathbf{d}$. For one active constraint the Karush–Kuhn–Tucker conditions give the solver-free closed form

$$\mathbf{u}^* = \mathbf{u}_{\text{nom}} + \frac{[-(L_{f_c} h + L_{g_c} h \mathbf{u}_{\text{nom}} + \alpha(h) - \kappa_d)]_+}{\|L_{g_c} h\|^2} (L_{g_c} h)^\top, \quad (5)$$

with $[\cdot]_+ = \max(0, \cdot)$, so the clamp runs in bounded, deterministic time inside the control loop and never invokes an iterative solver on the hot path.

3.3 Forward invariance (the guarantee)

Proposition 1 (Forward invariance). *Let $\alpha(h) = \gamma h$ with $\gamma > 0$, and suppose $\mathbf{u}^*$ from (4) is applied. Then along any solution, $\dot{h} \geq -\gamma h$, hence*

$$h(\mathbf{x}(t)) \geq h(\mathbf{x}(0)) e^{-\gamma t} \geq 0 \quad \text{whenever } h(\mathbf{x}(0)) \geq 0, \quad (6)$$

so $\mathcal{C}$ is forward-invariant: once inside the safe set the state cannot leave it.

Proof. By construction $\mathbf{u}^*$ satisfies (4), so $\dot{h} = L_{f_c} h + L_{g_c} h \mathbf{u}^* \geq -\gamma h + \kappa_d - |\nabla h \cdot \mathbf{d}| \geq -\gamma h$ under $\|\mathbf{d}\| \leq \bar{d}$. Grönwall's inequality applied to $\dot{h} \geq -\gamma h$ yields (6). On the boundary $h = 0$, $\dot{h} \geq 0$, so the field points into $\mathcal{C}$ (Nagumo's condition), which is exactly forward invariance. $\square$

Proposition 1 converts “we never observed a violation” into “it provably cannot leave $\mathcal{C}$ under the stated disturbance bound.” The QP feasibility of (4) is guaranteed whenever $L_{g_c} h \neq 0$ (relative degree one); relative-degree-two barriers use the exponential-CBF extension. A stronger whole-reachable-set statement — $\text{Reach}(\mathcal{X}_0) \cap \mathcal{X}_{\text{unsafe}} = \emptyset$ via Hamilton–Jacobi reachability with neural-network verification — is the assurance-grade successor on the roadmap.

3.4 Closed-loop stability under surrogate error

The interconnection of the performance controller (1) and the clamp (4) is input-to-state stable (ISS) with respect to the surrogate error $\mathbf{e} = f - \hat{f}$: if the nominal MPC admits a Lyapunov function V with $\dot{V} \leq -c_1 \|\mathbf{x} - \mathbf{x}^*\|^2$ and the surrogate error is bounded $\|\mathbf{e}\| \leq \bar{e}$, then the clamped closed loop satisfies $\dot{V} \leq -c_1 \|\mathbf{x} - \mathbf{x}^*\|^2 + c_2 \bar{e}$ inside $\mathcal{C}$, so trajectories converge to a residual set of size $\mathcal{O}(\bar{e})$ while never leaving $\mathcal{C}$. Safety (Proposition 1) is thus *unconditional* in $\bar{e}$; only performance degrades gracefully as the model worsens. This separation — hard safety, soft performance — is the reason a learned controller can be trusted in a nuclear loop.

3.5 Demonstrated result

The clamp is exercised on the breeder pressure limit (gate KX-L1-A13). With the learned controller alone the peak normalized pressure reaches $\beta_N = 4.200$ and violates the $\beta_N \leq 3.5$ safe set in 50 of 100 Monte-Carlo trajectories; with the CBF projection the peak sits exactly on the limit ($\beta_N = 3.500$), the minimum barrier value is $h_{\min} = +0.000$, and 0 of 100 trajectories escape, with the control effort held in $\mathbf{u} \in [1.06, 1.25]$ (figure 4). Forward invariance holds numerically as Proposition 1 predicts. The certified-filter study, including the honest PINN negative discussed in §4, is a dedicated companion paper.

Control-barrier safety projection (KX-L1-A13): $\mathbf{u} \in [1.06, 1.25]$

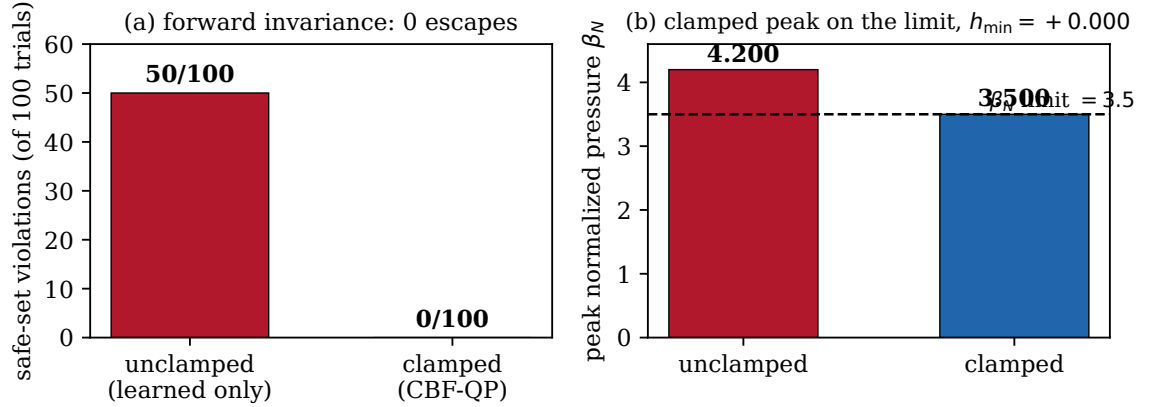


Figure 4. Control-barrier safety projection (KX-L1-A13). (a) Safe-set violations over 100 trajectories: 50 unclamped versus 0 clamped. (b) Peak β_N : the learned controller reaches 4.200; the clamped controller sits on the $\beta_N = 3.5$ limit with $h_{\min} = +0.000$. Values are the frozen register anchors; no trajectory data are invented.

4 Learned surrogates and calibrated uncertainty

4.1 Physics-informed equilibrium reconstruction

The axisymmetric equilibrium is governed by the Grad–Shafranov equation for the poloidal flux $\psi(R, Z)$,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi), \quad (7)$$

with pressure $p(\psi)$ and poloidal-current function $F(\psi) = RB_\phi$. A PINN ψ_θ minimises a composite loss that embeds (7) as a residual at collocation points,

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{data}}}_{\text{fit}} + \lambda_{\text{pde}} \frac{1}{N} \sum_{i=1}^N \|\mathcal{N}[\psi_\theta](\mathbf{x}_i)\|^2 + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}}, \quad \mathcal{N}[\psi] \equiv \Delta^* \psi + \mu_0 R^2 p'(\psi) + F F'(\psi), \quad (8)$$

where $\lambda_{\text{pde}}, \lambda_{\text{bc}}$ trade data fit against physical consistency and boundary conditions [16, 17].

Against the free-boundary solver FreeGS/FreeGSNKE [23] the trained network reaches an RMS- ψ error of 0.139% — clearing the 1% acceptance target with margin, against a 1.79% prior baseline — with a Grad–Shafranov operator residual of 0.32%, at a $2877\times$ speed-up over the iterative solver (gate KX-L1-A11, figure 5). The differentiability of ψ_θ is what lets the controller linearise the twin on the fly (§3).

4.2 Neural-operator transport surrogates

The controller needs the turbulent fluxes at latency. Let a denote the local plasma state (normalised gradients $R/L_T, R/L_n$, safety factor, shear, collisionality) and $u = \{Q_i, Q_e, \Gamma\}$ the heat and particle fluxes; the nonlinear gyrokinetic map $u = \mathcal{G}(a)$ is defined by CGYRO [21] (cf. GENE [22]). We emulate it by a neural operator $\mathcal{G}_\theta$ in the DeepONet/Fourier-neural-operator family [14, 15]: DeepONet factorises the operator into a branch network of the input function and a trunk network of the evaluation coordinate, $\mathcal{G}_\theta(a)(y) \approx \sum_{k=1}^p b_k(a) t_k(y)$, while the FNO parameterises the kernel in the spectral domain, $(\mathcal{K}v)(x) = \mathcal{F}^{-1}(R_\phi \cdot \mathcal{F}v)(x)$ with learned low-mode weights R_ϕ . The surrogate is accepted only when its held-out error clears

$$\text{NRMSE}(\mathcal{G}_\theta) = \frac{\left[\frac{1}{N} \sum_k \|\mathcal{G}_\theta(a_k) - u_k\|^2 \right]^{1/2}}{u_{\max} - u_{\min}} < 5\% \quad (\text{gate SH-026}), \quad (9)$$

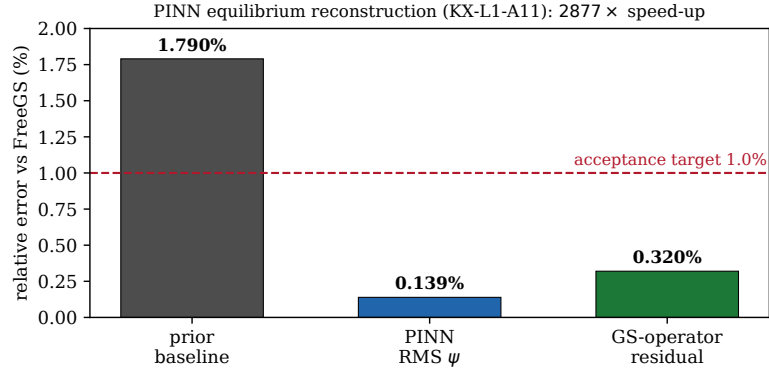


Figure 5. PINN equilibrium reconstruction (KX-L1-A11): RMS- ψ error 0.139% against the 1% target (prior baseline 1.79%), Grad-Shafranov residual 0.32%, at $2877\times$ speed-up over FreeGS. Frozen register values.

and the equilibrium operator clears $\|\mathcal{E}_\phi - \psi_{\text{ref}}\|_2 / \|\psi_{\text{ref}}\|_2 < 0.5\%$ against a free-boundary solver (SH-031). These surrogates, and the manifold-abstention monitor that flags an out-of-envelope operator call, are the subject of a companion neural-operator paper.

4.3 Graph-network state reconstruction and robustness

Diagnostics degrade under a 14 MeV neutron field, so the state estimate must survive sensor loss. A GNN over the diagnostic graph refreshes each node from itself and its physical neighbours,

$$\mathbf{h}_v^{(\ell+1)} = \sigma\left(\mathbf{W}_s \mathbf{h}_v^{(\ell)} + \sum_{u \in \mathcal{N}(v)} \mathbf{W}_n \mathbf{h}_u^{(\ell)}\right), \quad (10)$$

so a dropped channel is reconstructed rather than lost (gate KX-L1-A12: reconstruction AUC 0.980, NRMSE 0.396 on synthetic data). Robustness is quantified end to end on the magnet-protection function: a fused strain-rate+magnetization quench-precursor detector reaches an area under the receiver-operating characteristic (ROC) of 0.992 — against 0.976 for magnetization and 0.921 for strain-rate alone — with true-positive rates of 0.857 and 0.597 at the 1% and 0.1% false-positive operating points that a protection trip demands (gate KX-L1-A20, figure 6). Under model-fusion the protective function retains a precursor AUC ≥ 0.95 even with the in-winding nitrogen-vacancy (NV) magnetometer channel degraded by up to $39\times$ (gate SH-037, figure 7) — so the safety function survives a heavily degraded sensor.

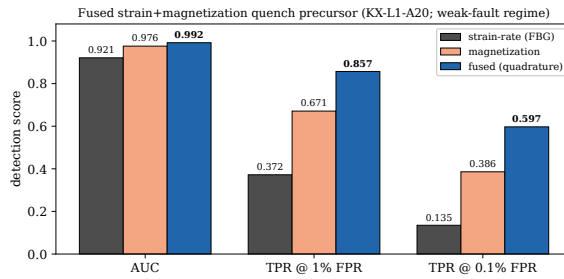


Figure 6. Fused quench-precursor detector operating points (KX-L1-A20), in a deliberately weak-fault regime. The fused channel dominates either single channel at AUC and at both low-false-positive operating points.

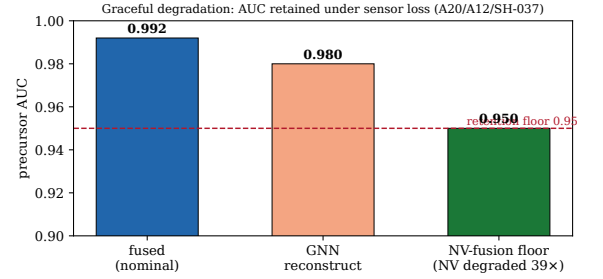


Figure 7. Graceful degradation (A20/A12/SH-037): the fused detector (0.992), GNN reconstruction (0.980), and the model-fusion floor that retains AUC ≥ 0.95 with the NV channel degraded $39\times$.

4.4 Ensemble assimilation and calibrated abstention

The shadow is an ensemble, so a calibrated distribution — not a best guess — reaches the safety layer. Given a forecast ensemble with mean $\bar{\mathbf{x}}^f$ and covariance $\mathbf{P}^f$, observations $\mathbf{y}$ with operator $\mathbf{H}$ and noise $\mathbf{R}$, the ensemble-Kalman update [24, 25, 26] advances each member by

$$\mathbf{K} = \mathbf{P}^f \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1}, \quad \mathbf{x}^{a,(m)} = \mathbf{x}^{f,(m)} + \mathbf{K}(\mathbf{y} + \boldsymbol{\epsilon}^{(m)} - \mathbf{H} \mathbf{x}^{f,(m)}), \quad \mathbf{P}^a = (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}^f. \quad (11)$$

Each model emits a mean and a variance, $\hat{u} = \mathbb{E}[u | \mathbf{x}, \theta]$ and $\text{Var}[u | \mathbf{x}, \theta]$, and the decision is *gated*: if the predictive variance exceeds a threshold, or a conformal/OOD monitor fires, authority is handed back to the deterministic floor,

$$\text{act on } \hat{u} \text{ iff } \text{Var}[u | \mathbf{x}, \theta] \leq \tau \wedge \mathbf{x} \in \hat{\mathcal{D}}_{\text{train}}, \quad \text{else abstain to rules/human.} \quad (12)$$

Conformal prediction supplies distribution-free, finite-sample coverage on every prediction — a nonconformity score calibrated on held-out data yields a set $C(\mathbf{x})$ with $\Pr(u \in C(\mathbf{x})) \geq 1 - \alpha$ regardless of the model [30] — and empirical coverage is accepted only if $|\text{Cov}_{\text{emp}}(\alpha) - \alpha| \leq 3\%$ (SH-027/073). OOD is detected as a Mahalanobis excursion from the training manifold and throttles model authority through the extrapolation ledger. Calibrated uncertainty, conformal wrapping, OOD gating, safe RL from operator feedback confined to the CBF-certified safe set [10, 11], and the signed-ledger runtime assurance each have a dedicated companion paper.

4.5 The acceptance contract

No component is granted control authority by assertion. Each clears the pre-registered inequalities of table 1 before it acts; these are the contract that gates the architecture (figure 3).

Table 1. The acceptance contract: pre-registered inequalities that gate control authority. Serials refer to the register [1]. “Demonstrated” denotes a frozen result; “target” a pre-registered bar not yet measured.

criterion	bar (serial)	status
CBF forward invariance	0 safe-set escapes, $h_{\min} \geq 0$ (KX-L1-A13)	demonstrated (0/100)
PINN equilibrium error	RMS $\psi < 1\%$ vs FreeGS (KX-L1-A11)	demonstrated (0.139%)
flux-operator error	NRMSE $< 5\%$ vs held-out CGYRO (SH-026)	target
equilibrium operator	RMS $\psi < 0.5\%$ at kHz (SH-031)	target
fused precursor AUC	dominate single channels (KX-L1-A20)	demonstrated (0.992)
sensor-loss retention	precursor AUC ≥ 0.95 , NV 39 $\times$ (SH-037)	demonstrated
ensemble coverage	$ \text{Cov}_{\text{emp}} - \alpha \leq 3\%$ (SH-027/073)	target
differentiable twin	AD vs FD Jacobian $< 10^{-3}$ (SH-032)	target
reduced-order fidelity	ROM error $< 5\%$ at $\geq 100\times$ (SH-093)	target
shadow lead-time	$\tau_{\text{lead}} \geq 50$ ms (all twins)	target

5 Computational methodology: the NVIDIA GPU HPC campaign

The programme is deliberately verification-first, and its compute is described here in full — codes, acceleration, scale and reproducibility — without recourse to any economic framing.

5.1 Reference codes and the de-risking campaign

Every headline number originates from a frozen reference code, not from the surrogate that later emulates it. The physics backbone spans nonlinear gyrokinetics (CGYRO [21], GENE [22]), extended-MHD stability, edge/divertor transport, Monte-Carlo neutronics and depletion (OpenMC [42]), fast-ion and alpha confinement, quasilinear transport, free-boundary equilibrium (FreeGS/FreeGSNKE [23]), and first-principles materials (DFT with machine-learned interatomic potentials). The de-risking campaign ran a large catalog of jobs to closure and stamped each as a frozen anchor in the register [1]; the surrogates of §4 are trained *against* these codes and admitted only when they clear the acceptance inequalities.

5.2 GPU acceleration and the three-clock partition

Training and large offline Monte-Carlo run on the NVIDIA GPU HPC campaign (A100/H100-class silicon); the online digital twin and model inference run on local GPU nodes at the millisecond tier, co-located with the plant for latency; and the hard real-time loop — the compute that reaches an actuator — runs on deterministic bare-metal and FPGA nodes with no hypervisor and no garbage collector. The three tiers are physically separated by role and by security domain and joined only by a physical data diode, so the microsecond protective loop never contends for bandwidth with a model download or a training job. A deterministic, time-sensitive network with sub-microsecond time distribution (PTP/White-Rabbit) gives every node and sensor a common timebase — the precondition for the latency-split clamp of (5) and for lineage timestamps to mean anything.

5.3 Reproducibility, verification, and MLOps

The pipeline is built for byte-level reproducibility. Every model is versioned with its data and configuration and promoted only through a validation gate — Tier-1 byte-exact reproduction and

Tier-2 tolerance reproduction against an independent library — and a model registry tracks what is deployed where, at which version, with which validator. Every training example carries its provenance (which pulse, which simulation, which device), and provenance-gated calibration is load-bearing for safety, not paperwork: control-barrier invariance held in 10/10 provenance-gated trials against 1/10 ungated. Every runtime decision is written to a signed, hash-chained ledger. No result in this paper depends on unpublished or proprietary data, and each is regenerable from the archived scripts and the register.

6 The resource-honest quantum frontier

We place quantum technologies where they actually are for fusion, and report the negatives.

6.1 Quantum sensing (Tier 2, near-term)

Magnetometry at or near the quantum limit sharpens the state estimate. The in-winding ensemble NV-diamond magnetometer reaches a shot-noise-limited sensitivity of $0.189 \text{ pT Hz}^{-1/2}$ at 125 kHz bandwidth (gate KX-L1-A22) [31, 32, 33]; a quench that sheds the screening current of a no-insulation turn produces a $\sim 100 \text{ mT}$ local step, roughly 10^9 times the per-shot floor, so the precursor is single-shot detectable with margin to spare for multiplexing and in-service degradation. Entanglement-enhanced sensing, however, helps only above a noise knee at $\sigma \approx 0.14$; all Kronos diagnostics sit below it, where the measured gain is $+0.00077 \text{ AUC}$ (gate BR-SX-08) — a near-null, kept in the record.

6.2 Quantum optimisation and reinforcement learning (Tier 3)

Casting actuator scheduling as a QUBO and solving it by QAOA or annealing shows *no measured quantum advantage*: classical simulated annealing dominates all three breeder QUBOs to $n = 120$ variables, the exact-to-heuristic crossover is at only $n = 12\text{--}16$, and QAOA plateaus at an approximation ratio of 0.90–0.95 and is classically unsimulable past ~ 24 qubits (gate BR-SX-06, figure 8). Barren-plateau scaling [37] and the general limitations of variational algorithms [36, 35, 41] are the mechanism, not an implementation defect. Quantum RL for shape control is on the roadmap, benchmarked against the classical RL controllers of §4, not claimed ahead of them.

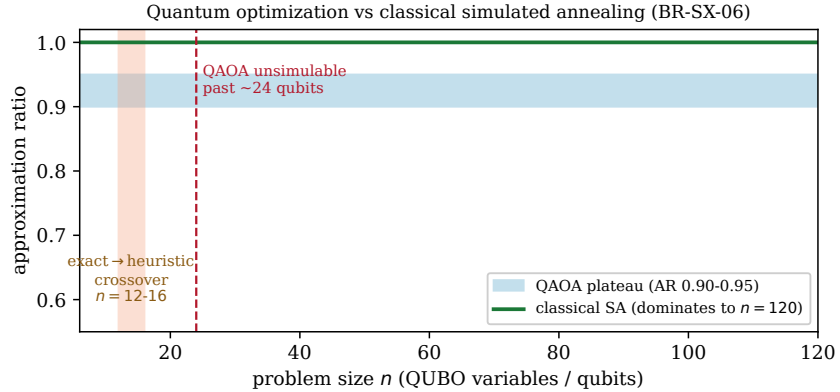


Figure 8. Quantum optimisation versus classical simulated annealing (BR-SX-06): QAOA plateaus at approximation ratio 0.90–0.95 while classical SA dominates to $n = 120$; the exact-to-heuristic crossover is at $n = 12\text{--}16$ and QAOA is unsimulable past ~ 24 qubits. No measured advantage.

6.3 Quantum machine-learning surrogates (a measured negative)

A quantum-kernel classifier surrogate does not improve with scale — it *degrades*: its AUC falls from 0.817 at 4 qubits to 0.691 at 8 qubits (gate BR-SX-08, figure 9), the signature of exponential kernel concentration in a widening feature map. We report this as a negative rather than reframing it; a fault-tolerant kernel would be 9–11 orders of magnitude slower than the classical baseline for this problem, so it is “never, for this problem,” not “soon.”

6.4 Fault-tolerant resource estimates and a dated roadmap

Where quantum computation could eventually help is offline calibration, and the honest resource split is load-bearing (figure 10). The linear kinetic (Vlasov) operator needs 11–19 logical qubits — early fault-tolerant — while the 6-D kinetic Hamiltonian simulation needs ~ 44 logical qubits and

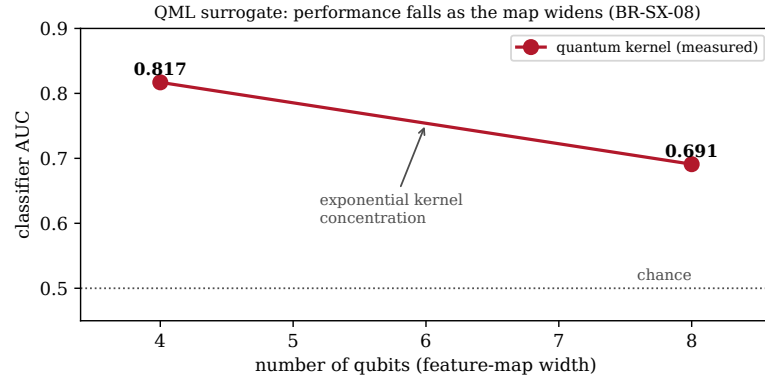


Figure 9. Quantum-machine-learning surrogate (BR-SX-08): the classifier AUC falls as the feature map widens from 4 to 8 qubits — exponential kernel concentration, reported as a measured negative.

~ 19 days, and exact alloy ground states by quantum phase estimation need ~ 238 logical qubits and ~ 38 days (gates KX-L3-A6/A1/A2), consistent with the finding that exponential advantage in ground-state chemistry is not established [40]. Resource estimates use the audited relations $N_{\text{iter}} = 1.6\sqrt{2^n}$ and $N_{\text{phys}} = N_{\text{logical}} \times 2d^2$ (HHL-class linear solvers [38]; variational chemistry [39]), recomputed with 0 failures across the estimate catalog. The dated roadmap (gate KX-L3-A7, figure 11) places early-fault-tolerant linear kinetics at $\sim 2029 \pm 2$, nonlinear 5-D gyrokinetics — the problem that would most help the machine — post-2035, and a fleet-scale Grover search at “never” (1.8×10^{18} iterations). Noisy hardware today shows no advantage; the classical baseline is cheaper at every demonstrated scale.

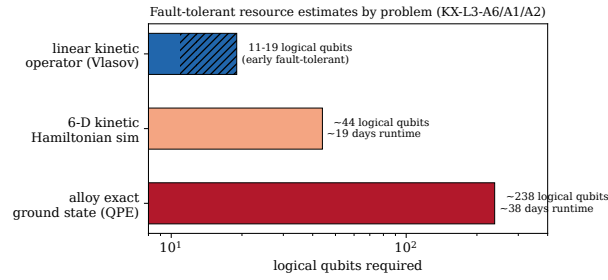


Figure 10. Fault-tolerant resource estimates by problem (KX-L3-A6/A1/A2): 11–19 logical qubits (linear kinetics), ~ 44 (6-D Hamiltonian), ~ 238 (alloy QPE). Log scale.

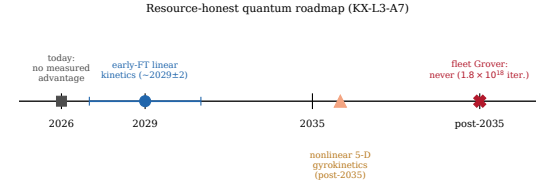


Figure 11. Dated quantum roadmap (KX-L3-A7): early-FT linear kinetics $\sim 2029 \pm 2$, nonlinear 5-D gyrokinetics post-2035, fleet Grover never.

7 Results

Table 2 lists the canonical design anchors both machines are held to; table 3 collects the Tier-1 AI results demonstrated in this paper; and table 4 collects the quantum resource estimates and negatives. The two magnets — the breeder centrepot at 16.84 T and the burner plug at 26.49 T — are modelled as the distinct devices they are throughout.

Table 2. Canonical design anchors (Tier-2 freeze). The AI/ML/quantum results are anchored to these frozen values; every model is verified against them.

quantity	Hyperion breeder	D- ³ He burner
gain	$Q = 3.076$	$Q_E = 1.318$
fusion power	$P_{\text{fus}} = 85.04 \text{ MW}$	(test $\rightarrow$ fleet)
plasma current	$I_p = 9.66 \text{ MA}$	—
triangularity	$\delta = -0.30$	—
peak magnet field	centrepot 16.84 T	plug 26.49 T
breeding / neutron fraction	TBR = 1.42	$f_n = 5.44\%$
plug-density ratio / DEC	—	$n_{\text{plug}}/n_{\text{cell}} \approx 16$; $\eta_{\text{DEC}} = 0.49$

Table 3. Tier-1 physics-informed AI results demonstrated in this paper. Serials refer to the register [1].

capability	metric	value	serial
CBF safety projection	safe-set escapes (clamped / unclamped)	0/100 vs 50/100	KX-L1-A13
	peak β_N (clamped)	3.500 ($h_{\min} = +0.000$)	KX-L1-A13
PINN equilibrium	RMS ψ vs FreeGS (target 1%)	0.139%	KX-L1-A11
	speed-up over iterative solver	2877 $\times$	KX-L1-A11
GNN reconstruction	AUC / NRMSE (synthetic)	0.980 / 0.396	KX-L1-A12
fused quench precursor	AUC (fused / mag / strain)	0.992/0.976/0.921	KX-L1-A20
	TPR @ 1% / 0.1% FPR (fused)	0.857 / 0.597	KX-L1-A20
NV magnetometer	sensitivity / bandwidth	0.189 pT Hz ^{-1/2} / 125 kHz	KX-L1-A22
sensor-loss retention	precursor AUC, NV degraded 39 $\times$	≥ 0.95	SH-037

Table 4. Tier-3 quantum resource estimates and measured negatives. “Advantage” is measured against the classical baseline at demonstrated scale.

problem	resource / result	horizon	advantage
linear kinetic (Vlasov) operator	11–19 logical qubits	early-FT ($\sim 2029 \pm 2$)	none today
6-D kinetic Hamiltonian sim	~ 44 logical qubits, ~ 19 d	frontier	none today
alloy exact ground state (QPE)	~ 238 logical qubits, ~ 38 d	frontier	none today
nonlinear 5-D gyrokinetics	fault-tolerant horizon	post-2035	none
QUBO actuator scheduling (QAOA)	AR 0.90–0.95; SA dominates to $n = 120$	—	negative
QML surrogate	AUC 0.817 $\rightarrow$ 0.691 ($4 \rightarrow 8$ qubits)	—	negative
entanglement-enhanced sensing	+0.00077 AUC below $\sigma \approx 0.14$ knee	—	near-null
fleet Grover search	1.8×10^{18} iterations	never	none

8 Discussion

8.1 Relation to learned control

The architecture is complementary to, not a competitor of, learned control. Degraeve *et al.* demonstrated a deep-RL policy commanding TCV coil voltages directly from magnetics [10], and Seo *et al.* an RL agent that steers DIII-D clear of a tearing mode [11]; both are learned *policies* acting on the present measurement. Kates-Harbeck *et al.* showed deep networks forecast disruptions across machines [12]. The Kronos contribution is orthogonal: a *certified boundary* beneath any such policy. A learned or RL controller can pursue performance while Proposition 1 guarantees the machine cannot leave the safe set, and the two compose — a policy may act on the shadow’s predicted, uncertainty-annotated state with the ≥ 50 ms lead, confined by construction to the CBF-certified safe set. What the published RL-control literature does not supply, and this work does, is the unconditional safety guarantee, the deterministic hardware floor, and the maturity gate that binds authority to verified credibility.

8.2 Relation to quantum-for-fusion claims

Against the broader “quantum for fusion” narrative, our stance is deliberately conservative and evidence-led. The variational-algorithm literature already documents the obstacles — barren plateaus [37], the limits of near-term devices [35, 36], and the absence of demonstrated exponential advantage in ground-state chemistry [40]. Our measured negatives (QAOA plateau, QML kernel concentration, the near-null entanglement-sensing gain below the $\sigma \approx 0.14$ knee) are consistent with that literature and, we argue, are more useful to the field than an optimistic projection: they say precisely which fusion problems are early-fault-tolerant (linear kinetics, ~ 2029), which are frontier (6-D kinetics, alloy QPE), and which are never (fleet Grover). Quantum sensing is the one near-term win, and even there the advantage is bounded.

8.3 Why the separation matters

The value of the architecture is that its tiers are honestly separated and its safety layer is certifiable. Hard safety is unconditional in the surrogate error; performance degrades gracefully; quantum sits where it actually is. The result is a system a regulator and an operator can reason about, and a method other programmes can adopt.

9 Limitations

Two honest gates bound the claims. First, and load-bearing: *there is no measured quantum advantage for fusion today*. Every quantum result here is a simulator-scale demonstration, a resource estimate, or a reported negative; the classical baseline is cheaper at every demonstrated scale, and the earliest crossover (linear kinetics) is an offline-calibration target near 2029, not a

control-loop capability. Second, we report the pre-registered misses rather than reframing them: the physics-informed equilibrium network was intended, in one variant, to become a fast *native* solver and in that variant did not reach the sharper target, and the certified safety guarantee is for the reduced-order operating-envelope model under a declared disturbance bound — elevating it to a whole-plant guarantee requires the Hamilton–Jacobi reachability and hardware-in-the-loop confirmation on the roadmap. Keeping these negatives in the record is part of the method.

10 Conclusion

The Kronos computational programme is a certifiable, resource-honest system and method for compact-fusion control. Physics-informed AI is deployable now *because* it is deterministically clamped: an MPC controller acts inside a control-barrier safety projection whose forward invariance we prove (Proposition 1) and realise as a solver-free QP, catching 50/100 would-be safe-set violations and holding the pressure limit with $h_{\min} = +0.000$. The models beneath it are fast and faithful — a PINN equilibrium at 0.139% RMS ψ and $2877\times$, neural-operator transport surrogates, a GNN that survives sensor loss, and a fused quench precursor at AUC 0.992 — carry calibrated uncertainty, and are admitted to authority only through pre-registered acceptance inequalities. The whole campaign runs verification-first on the NVIDIA GPU HPC campaign. Quantum sensing is a near-term diagnostics enhancement; quantum computation is an offline calibration tier with no crossover this decade, and we report the QAOA and QML negatives rather than reframing them. Two distinct magnets — breeder centrepost 16.84 T, burner plug 26.49 T — are modelled as such throughout. The contribution is a system and a method, not new plasma physics; each result is a frozen anchor in the de-risking register and heads a family of companion papers in the Kronos 2026 series.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. Every headline quantity is a frozen anchor in the KRONOS Physics De-Risking Register [1] (doi:10.5281/zenodo.22133057) and is regenerable from the archived, uniquely-named figure-generation script deposited with this paper; this paper is archived at doi:10.5281/zenodo.22645702; official version: <https://www.kronosfusionenergy.com/publications/ai-quantum-control.html>. As the umbrella of the computational series, this paper’s results are developed in dedicated companion papers, cross-cited here by their reserved DOIs: the four coupled digital twins (doi:10.5281/zenodo.22645805); the certified control-barrier safety filter (doi:10.5281/zenodo.22645716); model-predictive burn control inside the safe envelope (doi:10.5281/zenodo.22645807); neural-operator flux surrogates (doi:10.5281/zenodo.22645803); sub-millisecond neural equilibrium reconstruction (doi:10.5281/zenodo.22645801); calibrated uncertainty, conformal prediction and abstention (doi:10.5281/zenodo.22645813); runtime assurance with signed-ledger provenance (doi:10.5281/zenodo.22645817); out-of-distribution detection and epistemic gating (doi:10.5281/zenodo.22645819); safe reinforcement learning from operator feedback (doi:10.5281/zenodo.22645823); maturity-gated actuation authority (doi:10.5281/zenodo.22645795); the diagnostic suite and GNN reconstructor (doi:10.5281/zenodo.22645943); quantum optimisation and reinforcement learning without claimed advantage (doi:10.5281/zenodo.22645903); quantum machine-learning surrogates, entanglement-enhanced magnetometry and fault-tolerant resource estimates (doi:10.5281/zenodo.22645905); and resource-honest quantum computing against the classical frontier (doi:10.5281/zenodo.22645718). No result depends on unpublished or proprietary data.

Scope — what this paper does not claim. The certified safety guarantee is for the reduced-order operating-envelope model under a declared disturbance bound, not a whole-plant guarantee; and quantum computation shows no measured hardware advantage for fusion this decade.

Data availability

The frozen values behind every figure and table are archived with this deposit (doi:10.5281/zenodo.22645702) and trace to the register [1] (doi:10.5281/zenodo.22133057) by the serial identifiers cited inline. The uniquely-named figure-generation script is included in the deposit. Any economic analysis is reported separately and is not part of the public record.

Competing interests

The author declares no competing financial interests.

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